

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
library(cowplot)
```

```
##  
## Attaching package: 'cowplot'  
##  
## The following object is masked from 'package:lubridate':  
##  
## stamp
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#read in nutrient/chem data  
PeterPaul.chem.nutrients <-  
  read.csv(  
    here("Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),  
    stringsAsFactors = T)  
  
# read in Niwot Ridge Litter data  
NIWO_Litter <-  
  read.csv(here("Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),  
    stringsAsFactors = T)  
  
#2  
  
#change data values to date rather than factor  
NIWO_Litter$collectDate <- ymd(NIWO_Litter$collectDate)  
PeterPaul.chem.nutrients$sampldate <- ymd(PeterPaul.chem.nutrients$sampldate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3  
my_theme <- theme_light()+theme(  
  plot.background = element_rect(  
    fill = "tan"),#makes the area around the plot tan
```

```

plot.title = element_text(
  color = "blue" #makes the title of the plot blue
)
)

```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the lm method. Adjust your axes to hide extreme values (hint: change the limits using xlim() and/or ylim()).

```

#4
PByPo4 <- PeterPaul.chem.nutrients %>%
  ggplot(aes(
    x = po4,
    y = tp_ug,
    color = lakename
  ))+
  #add title
  ggtitle("Phosphorus by Phosphate in Peter and Paul Lakes")+

  #make points more transparent so trend lines are more visible.
  geom_point(alpha = 0.5)+

  #add y and x limits that will omit outliers from the plot.
  ylim(0, 150) +
  xlim(0, 40)+

  #add lines of best fit
  geom_smooth(method = "lm", se = FALSE)+

  #add x and y labels
  ylab("Phosphorus (t_ug)") +
  xlab("Phosphate (po4)") +

  #change legend title
  labs(color = "Lake Name") +
  my_theme #call my theme

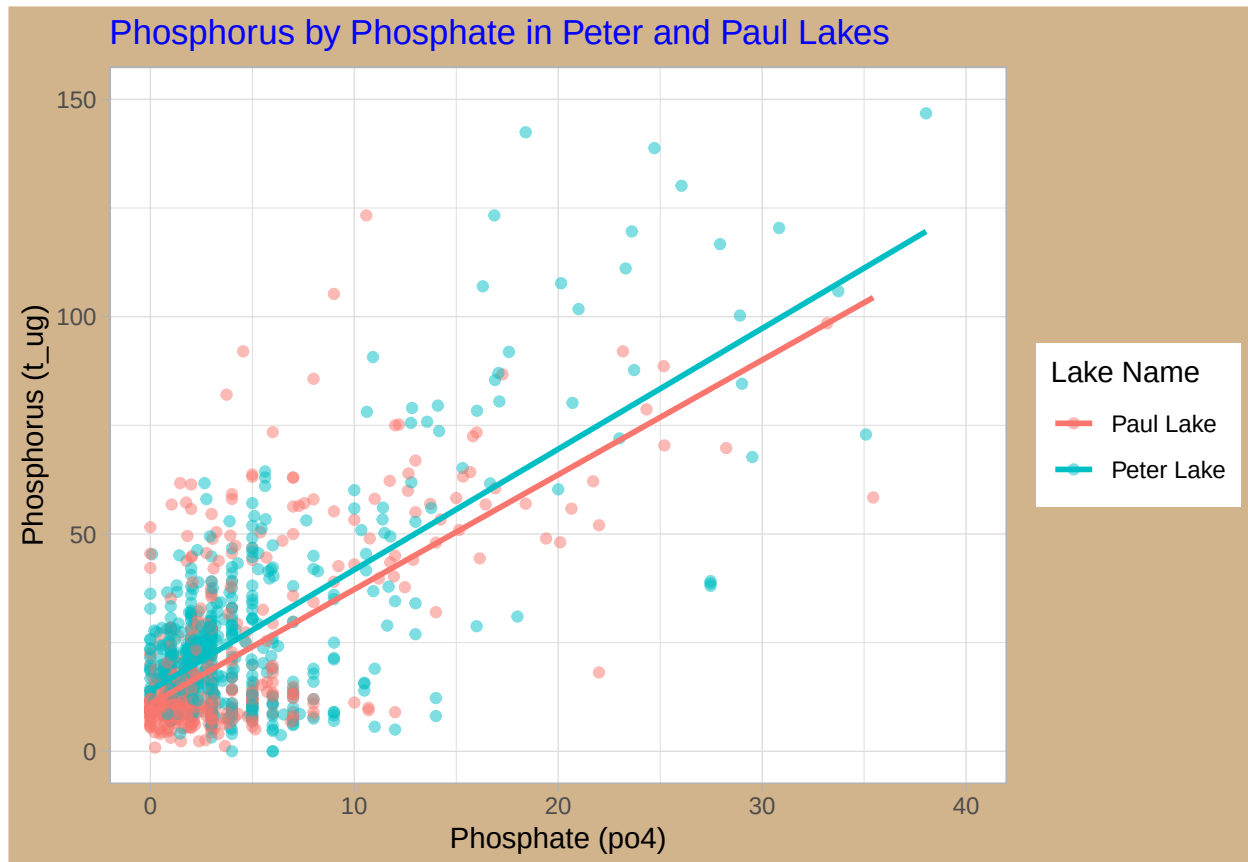
#plot the graph
PByPo4

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 21949 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 21949 rows containing missing values or values outside the scale range
## ('geom_point()').

```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
#create a box plot of temperature for each month
Temp_box <- PeterPaul.chem.nutrients %>%
  ggplot(
    aes (
      #treat month column as a factor (not a continuous variable) and label by
      #month name
      x=factor(month,levels = 1:12,labels = month.abb),
      y = temperature_C,
      color = lakename
    ))+
  xlab("Month")+
  ylab("Temp")+
  geom_boxplot()+
  #remove the x axis labels and legend
  theme(axis.text.x = element_blank(), axis.title.x = element_blank(), legend.position = 'none')
```

```

TP_box <- PeterPaul.chem.nutrients %>%
  ggplot(
    aes (
      x=factor(month,levels = 1:12,labels = month.abb),
      y = tp_ug,
      color = lakename
    ))+
  xlab("Month")+
  ylab("TP")+
  geom_boxplot()+
  #remove the x axis labels and legend
  theme(axis.text.x = element_blank(), axis.title.x = element_blank(),legend.position = 'none')

TN_box <- PeterPaul.chem.nutrients %>%
  ggplot(
    aes (
      x=factor(month,levels = 1:12,labels = month.abb),
      y = tn_ug,
      color = lakename
    ))+
  xlab("Month")+
  ylab("TN")+
  labs(color = "Lake Name")+
  #keep the x axis labels/title and legend but move the legend to the bottom
  theme(legend.position = 'bottom')+
  geom_boxplot()

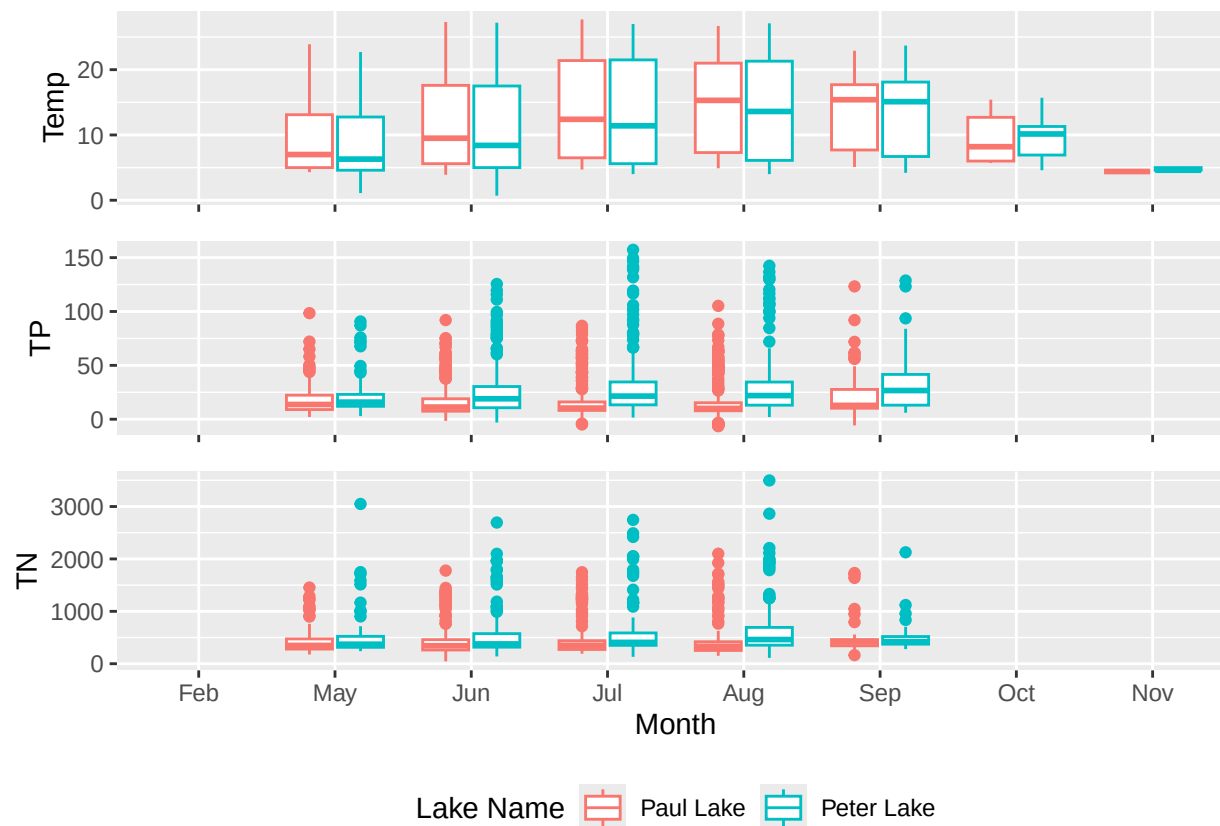
#create a cowplot of all three heights. Make the graphs aligns vertically and
#matched to the bottom plot's x axis. Because the bottom plot includes the x
#axis labels and key, make it's relative height larger so that the three graphs
#look approx the same height
plot_grid(Temp_box, TP_box, TN_box, rel_heights = c(1,1,1.75), nrow = 3, align = "v",axis = "tblb")

## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').

## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').

## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

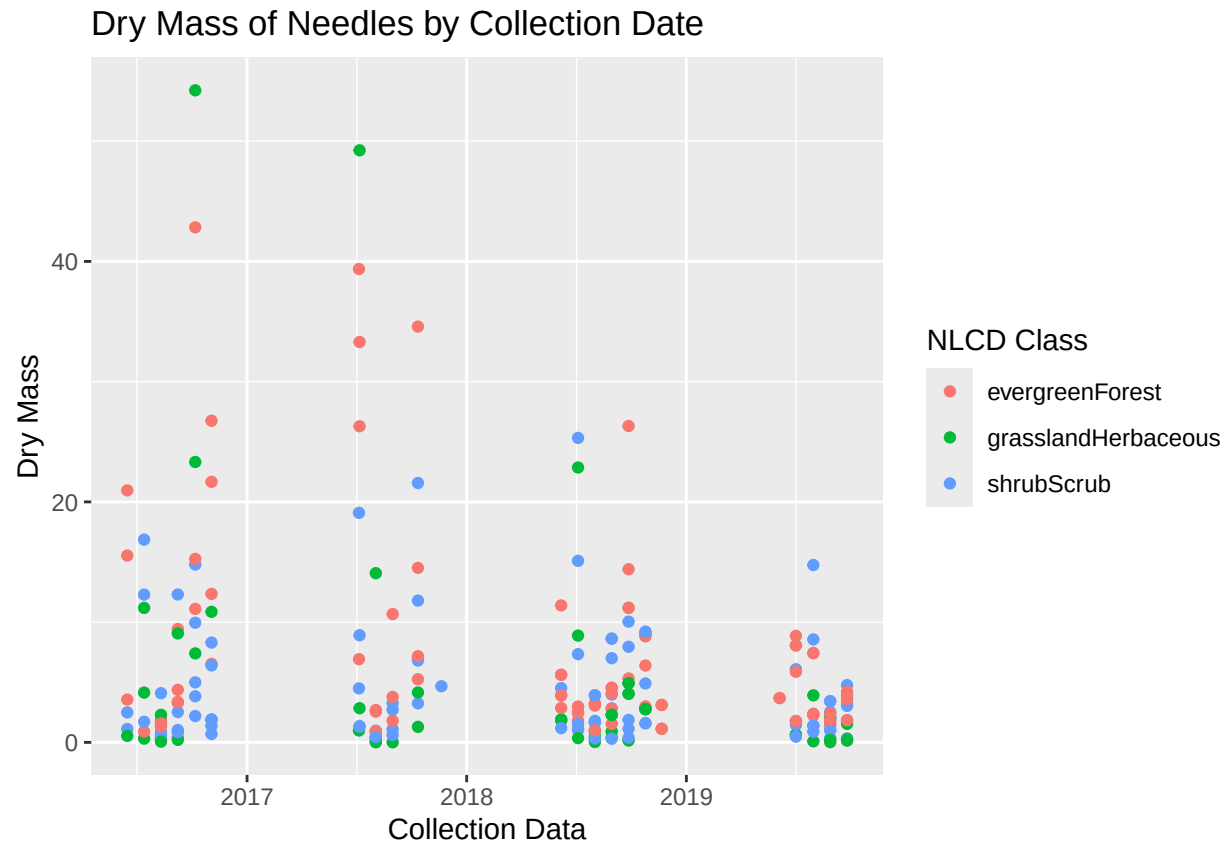


Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: All the variables (Temp, TP, TN) seem to be highest in the summer months (Jun, Jul, Aug) and lowest in the winter months (Nov, Jan Feb)

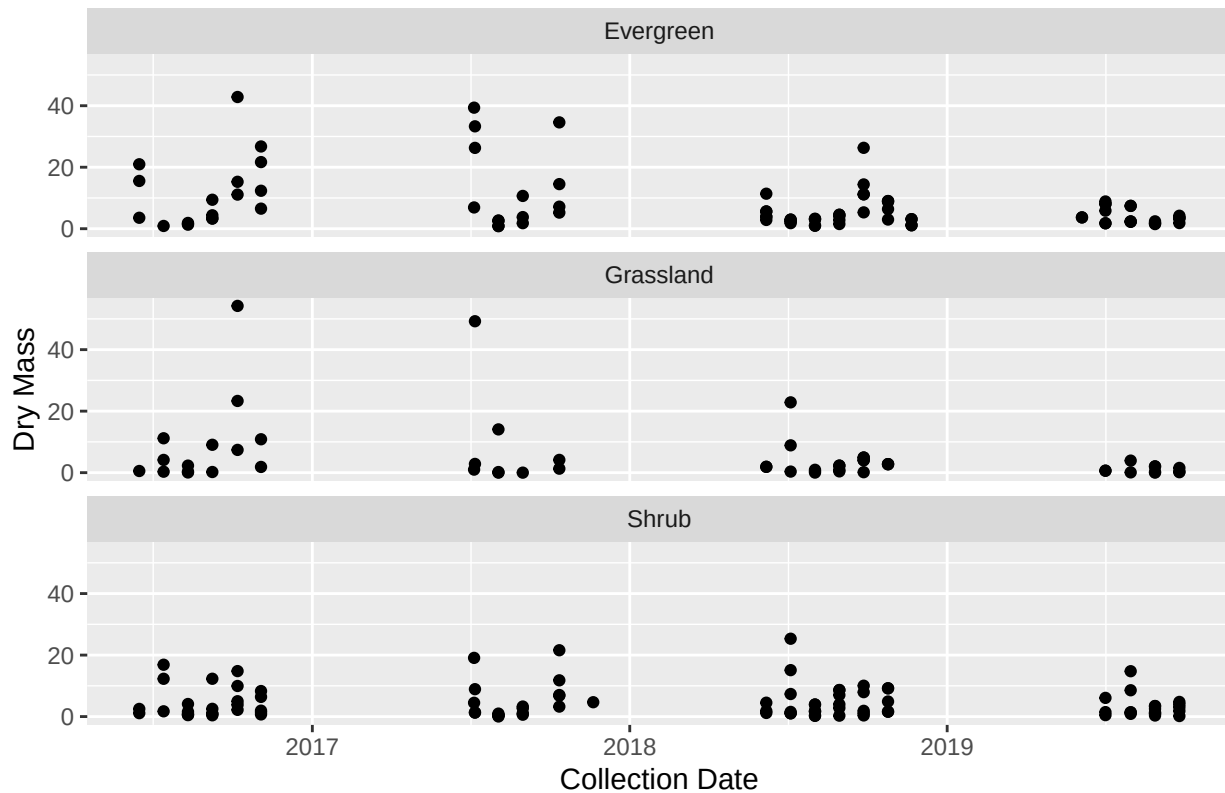
6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
Needle_Litter <- NIWO_Litter %>%
  filter(functionalGroup == "Needles") %>%
  ggplot(
    aes(
      x = collectDate,
      y = dryMass,
      color = nlcdClass
    ))+
  xlab("Collection Date")+
  ylab("Dry Mass")+
  ggtitle("Dry Mass of Needles by Collection Date")+
  labs(color = "NLCD Class")+
  geom_point()
Needle_Litter
```



```
#7
Needle_Litter_Facet <- NIWO_Litter %>%
  filter(functionalGroup == "Needles") %>%
  ggplot(
    aes(
      x = collectDate,
      y = dryMass
    ))+
  xlab("Collection Date")+
  ylab("Dry Mass")+
  ggtitle("Dry Mass of Needles by Collection Date")+
  geom_point()+
  #Use facet wrap to create a graph for each class, graphed along the same x axis. use labeller to chan
  facet_wrap(facet = vars(nlcdClass), nrow = 3, labeller = labeller(nlcdClass=c("evergreenForest" = "Ev
Needle_Litter_Facet
```

Dry Mass of Needles by Collection Date



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I think 7 is more effective because it is easier to see if there are trend differences among the three classes. 6 shows us that there is clearly a peak in all three classes around fall of each year, but it's not obvious if each class behaves similarly or differently within each fall. Changing the point size in 6 could help so we could more easily discern data points, but 7 very effectively shows the trend of dry mass within each type. In this example it doesn't matter too much since it doesn't seem like there is an obvious difference in dry mass trend among the different class types, but 7 helps us confirm this observation that there isn't much difference among the classes.